Early Detection of Pancreatic Cancer using Machine Learning Algorithm

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***Abstract*—Pancreatic cancer is very challenging since it has poor results when it is diagnosed at the later stages, and proper testing tools are lacking to treat patients. Endoscopic harmonic ultrasound imaging is used to characterize pancreatic masses. For detecting the pancreatic cancer machine learning techniques are mostly used. The main aim of this work is to compare and explain how NN and logistic algorithms give better results when it works with ensemble ML algorithms for diagnosing pancreatic cancer. The datasets are collected through the Kaggle platform. When compared to related work from the literature, it is shown that the ANN approach with a logistic algorithm achieved 90.50% accuracy from another machine learning algorithm.**

Keywords—Endoscopic, logistic, ANN, CNN, XgBoost, accuracy, Correlation matrix

I. INTRODUCTION

Pancreas is a digestive organ situated in the stomach. It is a rare disease that occurs fewer than 1 million people every year. For adults, it's 6 inches long and less than 2 inches wide. The head of the pancreas is fixed on right side of the abdomen. Th pancreas is fixed behind the lower stomach. Pancreas is the aggressive type of cancer which starts from the tissues of pancreas. Pancreas plays an important role in digestive system, which supply enzymes that helps in food digestion and hormones like insulin and glucagon, helps to regulate blood sugar levels. Pancreatic cancer initializes when the unusual cells within the pancreas grows and multiply out of hand, which forms a tumor. There is no sign of getting a disease in the early stage, but a later stage of disease it can be non-specific, such as lack of appetite and weight loss. The main cause of this disease is not able to detect any symptoms in the earlier stages, which makes it challenging to diagnose. As a result, pancreatic cancer is detected at an early stage to avoid high mortality rate.

The study suggested a way of using the imaging process, collecting various datasets from patients, computer aided diagnosis to address the issues based on images. The main risk factors that develop the pancreatic cancer are age, obesity, diabetes, family history. Although the subjects of the study include different stages of Pancreatic cancer, and there is a lack of future study of the normal pancreas and different stages of pancreatic cancer. The study contributes that the study has a certain staging of Pancreatic cancer based on CT images and diagnosis of patient's health factors. Also, to explore the early screening and staging of Pancreatic cancer by using machine learning algorithms to be used in the diagnosis process for their well-being.

II. RELATED WORKS

Artificial Neural Networks, or ANNs, have played a significant role in detection for particular tasks like involving the various data types like field of medical images, genetic data, and clinical data, including the prediction and diagnosis of cancer [3][4]. Typically, it uses the Image data Acquisition which involves the process of acquiring medical images of the pancreas. The images that are acquired are needed for detecting abnormalities and tumors in the pancreas. Using an ANN model, scientists and researchers were able to diagnose the cancer based on the symptoms. Input Layer: Neurons here receive the input data, each neuron representing a feature. Hidden Layers: Intermediate layers process the input through weighted connections, applying transformations to recognize patterns. Output Layer: This layer produces the final output, such as the predicted class (healthy or affected by pancreatic cancer). ANNs learn by adjusting the weights assigned to connections between neurons during training. This learning process involves feeding data through the network, comparing the predicted output to the actual output, and adjusting weights to minimize the difference (error) between them. The logistic algorithm is often used as the activation function in the output layer of neural networks for binary classification tasks (such as healthy vs. cancer).

Using the training dataset, 10-fold stratified cross validation is used to find the performance of the models trained on the various datasets. The output of an ANN is a fractional number which varies between 0 and 1. The higher output value means the higher risk of pancreatic cancer. Also, the performance has both positive and negative status of the evaluation result.

III. THE XGBOOST ALGORITHM

XgBoost stands for "Extreme Gradient Boosting" and it is one of the most popular and widely used machine learning algorithms due to its ability to handle large datasets and its ability to achieve better accuracy and performance in ML tasks such as classification and regression. Also, XgBoost has built-in-support for parallel processing.

A distributed gradient boosting library optimized for its efficiency and scalability in machine learning model training is called XgBoost. It is an ensemble learning techniques that gives a stronger prediction by aggregating the predictions of several weak models. The key components of the XgBoost algorithm are as follows:

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A. The Framework for Gradient Boosting:

XgBoost is a fast and efficient implementation of gradient boosting machines. Gradient boosting is an ensemble learning strategy in which weak learners, typically decision trees, are trained one after the other in an attempt to fix each other's mistakes..

B. Regularization:

To prevent overfitting, XgBoost incorporates regularization terms into its goal function. By doing this, it is possible to avoid the model fitting the training data too closely and having poor performance on fresh, untested data. Identify applicable funding agency here. If none, delete this text box.

C. Tree Pruning:

Pruning is employed to remove splits that don't contribute significantly to reducing the loss. This helps in controlling the complexity of the trees and prevents overfitting.

D. Parallel Processing:

The scalability and efficiency of XgBoost are built in. It is quicker than many other gradient boosting implementations because it can take use of parallel computing.

E. Handling Missing Data:

XgBoost comes with built-in tools to deal with missing data. During training, it gives missing data a direction, and it maintains this orientation throughout predictions.

F. Cross-validation:

To aid with model tuning, XgBoost has a cross-validation feature. This helps avoid overfitting and enables hyperparameter tweaking.

Train-Validation Split: Split the dataset into separate training and validation sets. Train the model on the training set and validate it on the separate validation set. This ensures that the model is evaluated on data it hasn't seen during training.

G. Feature Importance:

The feature importance score that XgBoost offers shows how important each feature is in relation to the other when it comes to producing predictions. Selecting features and comprehending the behavior of the model can both benefit from this.

H. Ensemble Methods:

Ensemble Methods: Techniques like bagging (Bootstrap Aggregating) and boosting (like XGBoost) combine multiple models to improve accuracy and prevent overfitting by reducing variance.

I. Early Stopping:

Monitor the model's performance on a validation set during training. Stop training when the model's performance on the validation set starts to degrade, indicating overfitting.

I. Dropout (Neural Networks):

In neural networks, dropout randomly deactivates some neurons during training, preventing any specific neuron from becoming overly reliant on certain features. Hyperparameter Tuning: Optimize hyperparameters using techniques like grid search or random search to find the best combination that minimizes overfitting.

IV. SYSTEM ARCHITECTURE

The Fig 1 represents the system architecture which comprises of front end and back end. The datasets that are collected through Kaggle (which is an open-source data set), which contains of relevant details like medical records, patient demographics, imaging data, etc., and then the data are preprocessed, and with the usage of the training algorithm, the deployment process will occur.

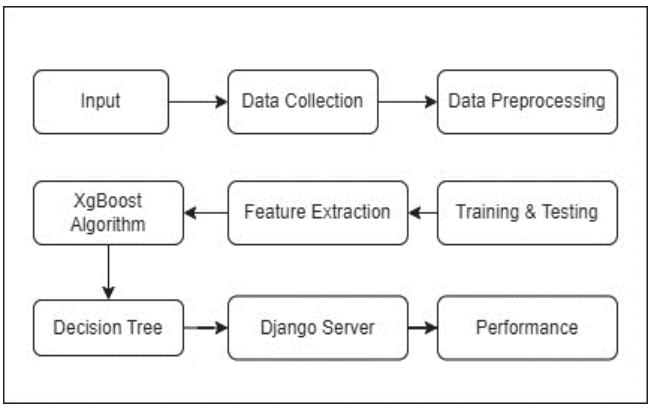

Fig. 1 Overall System Architecture

The frontend accepts the source of inputs and combines it with the backend, which is the server that processes the overall input that is given by the user and when the process gets completed, the resultant prediction is displayed.

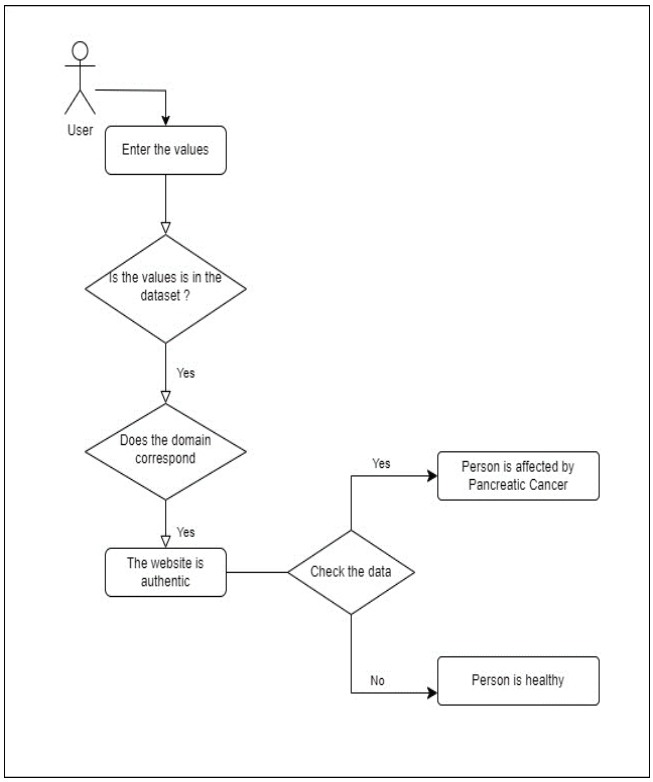

Fig. 2 Pancreatic Cancer Detection Architecture

The Fig 2 represents the architecture diagram where the input features are provided and then the prediction is done

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using the algorithms. If they are affected in early stage, it will detect the output as patient is affected by mild pancreatic cancer whereas if they are affected in advanced stages, it will detect the output as patient is leading to transplant of pancreas or it may lead to patient's death or another output that detects and displays is that the patient is a healthy person.

V. IMPLEMENTATION

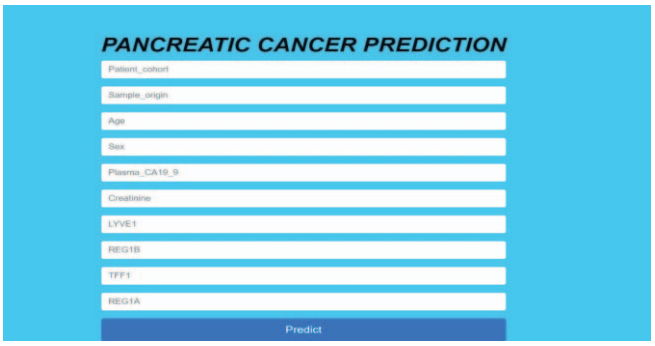

Fig. 3 Home Screen of pancreatic cancer application

Fig 3 shows the home screen of the web application, where it contains the details. In the home screen, the user should provide the features input. Also, the user should enter their input values to proceed further for detection.

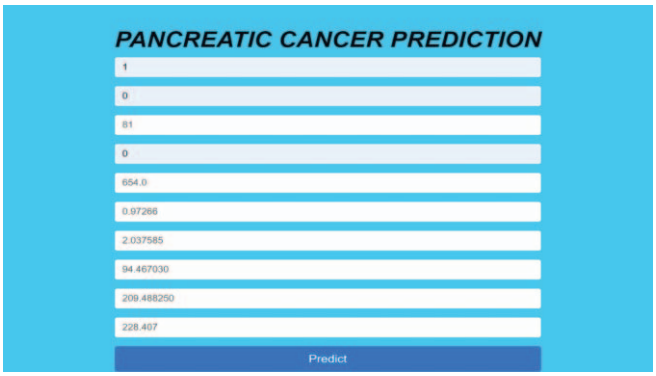

Fig. 4 entered input values in the application

Fig 4 shows the home screen of the web application when the required details are submitted. When submitted, the details will be sent to the backend server which does all the processing. The user has to wait until the processing gets completed.

Fig. 5 output

Fig 5 represents the detected output that the patient is healthy or the patient is affected by pancreatic cancer. This is how the web application looks like and this is the overview of the early detection of pancreatic cancer system.

VI. EXPERIMENT AND PERFORMANCE ANALYSIS

Below are the execution or data analysis on how the features inserted are detected and evaluated.

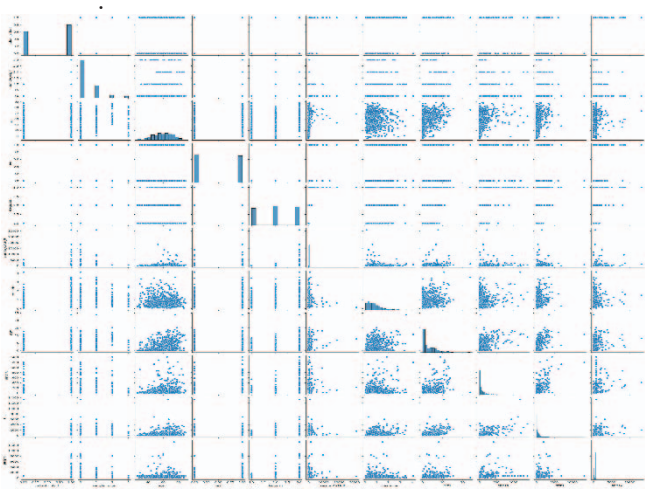

Fig. 6 Features used in Detection

Fig 6 represents the features that are used to calculate the detection, features such as Patient_cohort, Sample_origin, Age, Sex, Plasma_CA19_9, Creatinine, LYVE1, REG1B, TFF1, REG1A etc. are used as an models to detect the pancreatic cancer.

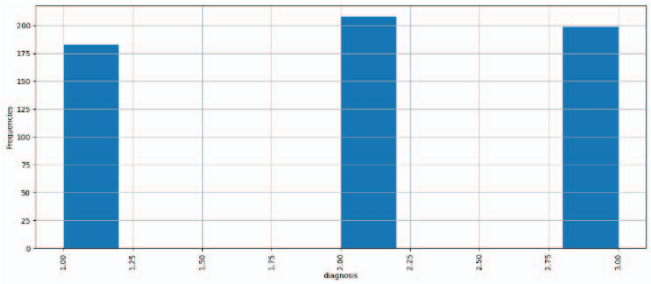

Fig. 7 Features Evaluation

Fig 7 is the detection measurement check on which the features does it gives more amount of detection. According to the above figure, the diagnosis rate gives the higher amount of diagnosis ranging from 1.0 to 3.0, so with the help of the frequencies range the detection can be done precisely.

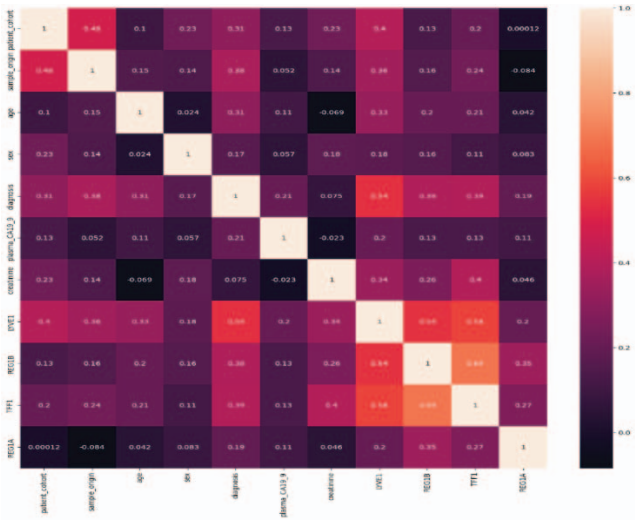

Fig. 8 Analysis Constraint Chart

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Fig 8 depicts the analysis constraint chart that ranges from 0.0 to 1.0. If any one of the features that is indicated in white, then it is denoted as patient is highly affected by pancreatic cancer or else the patient is mentioned as a healthy person. Also, these chart represent the performance of the pancreatic cancer and they are evaluated according to the detection.

VII. FUTURE WORK

Thus, the proposed system continues to explore and create new features that may provide more valuable information for the Cancer prediction. Collaboration with the domain experts can help us to identify relevant features and also by evaluating the cost-effectiveness of using various model for cancer prediction may be compared to the traditional diagnostic methods. In order to improve patient care and ease the diagnosis process, we may incorporate the model with Electronic Health Record (EHR) systems. By extracting better features from the model to keep up higher accuracy rate to detect the cancer prediction. By using combination of machine learning algorithm can give better work and can maintain perfect accuracy.

VIII. CONCLUSION

Thus, the overall research says that the pancreatic cancer is a serious disease known for its aggressive nature and challenging prognosis. The use of ML algorithms, can analyze vast and diverse datasets, which also apply them in clinical data, genomics, and medical imaging, can be analyzed by the algorithms to detect patterns, biomarkers, and relevant risk factors related to pancreatic cancer. This can lead to early detection, accurate diagnosis, and personalized treatment strategies tailored to the specific characteristics of each patient's cancer. As a result of using Early detection of pancreatic cancer using XgBoost an accuracy rate of 95.50% is achieved.

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